

# A Sharp Fixed-Defect Conjecture for Fractional Gaussian Traces

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September 5, 2026

## Abstract

The coefficientwise comparison between the integer traces of fractional Gaussian binomial coefficients at parameters  $1/2$  and  $1/4$  remains unresolved in the cited versions of Han–Xiong and Ono. We formulate a fixed-defect conjecture for the second increment of a finite Gaussian numerator. The proposed correction is a ten-term polynomial, and all ten coefficients are necessary already at  $k = 13$ . Two exact recurrences and a finite computational bridge show that the conjecture would imply the trace comparison. Archived integer computations support the conjecture for  $1 \leq k \leq 1004$  in every degree; independent reconstruction covers explicitly smaller ranges. Stabilization extends this evidence to every  $k$  in degrees at most 1005. We give exact falsification criteria, describe failed positivity routes, and distinguish the finite evidence from the unproved uniform bound.

## 1 The unresolved trace comparison

For  $k \geq 0$  put  $(a; q)_k = \prod_{i=0}^{k-1} (1 - aq^i)$  and

$$G_{r,k}(q) = \frac{(q^{r+1}; q)_k}{(q; q)_k}.$$

If  $r = a/b > 0$  is reduced, this is a formal series in  $\mathbb{Q}[[q^{1/b}]]$ . Its integer trace  $T_{r,k}$  retains precisely the terms with nonnegative integral exponent. For ordinary formal series,  $H \succeq_q 0$  means  $[q^n]H \geq 0$  for every integer  $n \geq 0$ . The problem considered here is

$$F_k := T_{1/2,k} - T_{1/4,k} \succeq_q 0 \quad (k \in \mathbb{Z}_{\geq 1}). \quad (1)$$

All comparisons are formal; no numerical value of  $q$  is prescribed. In particular, division by  $1 - q$  and by  $(q; q)_k$  is legitimate because their constant terms are one. Negative degrees and  $k = 0$  are outside (1).

Han and Xiong introduced the trace comparison in their fractional-index conjecture [3]. Ono’s support-dominance and canonical-reduction results settle other parameter ranges, but leave the full  $r = 1/4$  family explicitly open in Section 3.4, equation (31), of [4]. The same source reports exact verification for  $k \leq 200$ ; its Appendix A formalizes the reduction, not this remaining family. These statements describe the cited versions consulted on September 5, 2026.

For  $0 \leq s \leq k$ , define the Gaussian polynomial

$$\begin{bmatrix} k \\ s \end{bmatrix}_q = \frac{(q; q)_k}{(q; q)_s (q; q)_{k-s}},$$

and take it to be zero otherwise. The finite  $q$ -binomial expansion, or Ono's trace formula [4, Proposition 2.2], gives

$$D_k = \sum_{j=0}^{\lfloor k/2 \rfloor} q^{2j(j+1)} \begin{bmatrix} k \\ 2j \end{bmatrix}_q - \sum_{\ell=0}^{\lfloor k/4 \rfloor} q^{8\ell^2+3\ell} \begin{bmatrix} k \\ 4\ell \end{bmatrix}_q, \quad F_k = \frac{D_k}{(q; q)_k}. \quad (2)$$

Indeed, integer exponents in the two numerator expansions select subset sizes divisible by two and four respectively; their signs are positive. Thus  $D_k/(1-q) \succeq_q 0$  suffices for (1), since each remaining factor  $(1-q^i)^{-1}$  has nonnegative coefficients. Necessity of this sufficient criterion is not asserted. In particular, a negative prefix of  $D_k$  is not automatically a counterexample to (1). The small cases are  $D_1 = 0$ ,  $D_2 = q^4$ , and  $D_3 = q^4 + q^5 + q^6$ .

## 2 The fixed-defect conjecture

Define

$$Z_k(q) = \sum_{j=0}^{\lfloor k/2 \rfloor} q^{2j(j+1)} \begin{bmatrix} k \\ 2j \end{bmatrix}_q - \sum_{\substack{\ell \geq 0 \\ 4\ell+2 \leq k}} q^{8\ell^2+11\ell+3} \begin{bmatrix} k \\ 4\ell+2 \end{bmatrix}_q, \quad (3)$$

$$B(q) = q^5 + q^6 + 2q^7 + 2q^8 + 3q^9 + 3q^{10} + 4q^{11} + 3q^{12} + 3q^{13} + q^{14}. \quad (4)$$

**Conjecture 1** (Fixed negative defect). *For every  $k \in \mathbb{Z}_{\geq 1}$  and every  $n \in \mathbb{Z}_{\geq 0}$ ,*

$$[q^n] \left( \frac{Z_k(q)}{1-q} + B(q) \right) \geq 0, \quad (5)$$

where  $Z_k$  and  $B$  are exactly (3) and (4).

In words, for each positive integer  $k$ , the cumulative coefficient sums of the specified difference of Gaussian polynomials are nonnegative after adding the same ten prescribed corrections, independently of  $k$ . This is a conjecture about the second increment of a numerator, not a restatement of the trace inequality. We prove a sufficient implication below; no converse is known here.

### 2.1 Subset interpretation and exact tails

For  $b = 2, 4$  let  $P_{b,k}(x) = \prod_{i=1}^k (1 + x^{bi+1})$ . The identity

$$\sum_{\substack{S \subseteq \{1, \dots, k\} \\ |S|=s}} q^{\sum_{i \in S} i} = q^{s(s+1)/2} \begin{bmatrix} k \\ s \end{bmatrix}_q$$

shows that

$$[q^n] Z_k = [x^{2n}] P_{2,k} - [x^{4n+2}] P_{4,k}. \quad (6)$$

More explicitly, an even-sized subset of size  $s$  contributes positively at degree  $\sum S + s/2$ . A subset with  $s \equiv 2 \pmod{4}$  contributes negatively at degree  $\sum S + (s-2)/4$ . These weights yield exactly the exponents in (3). The empty subset contributes one, so  $Z_1 = 1$ . The same counting gives

$$Z_k(1) = \sum_{\substack{0 \leq s \leq k \\ s \equiv 0 \pmod{4}}} \binom{k}{s} > 0. \quad (7)$$

For each fixed  $k$ , prefixes beyond  $\deg Z_k$  are therefore this positive constant. Every finite-row check can cover all degrees, with no analytic tail estimate or floating-point tolerance.

## 2.2 Why the correction is sharp

Exhaustive subset enumeration at  $k = 13$  gives the following complete negative part of  $Z_{13}/(1-q)$ :

| $n$                 | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 |
|---------------------|----|----|----|----|----|----|----|----|----|----|
| $[q^n]Z_{13}/(1-q)$ | -1 | -1 | -2 | -2 | -3 | -3 | -4 | -3 | -3 | -1 |

All other prefixes in that row are nonnegative. The finite calculation is reproducible by the enumeration procedure in Section 4.2. Consequently every nonnegative polynomial  $\tilde{B}$  satisfying a uniform bound  $Z_k/(1-q) + \tilde{B} \succeq_q 0$  must obey  $\tilde{B} \succeq_q B$ . Through degree 14, only positive cardinalities 0, 2, 4 and negative cardinality 2 contribute. The corresponding rectangle coefficients have stabilized by  $k = 13$ : the largest required degree is 11 in  $\begin{bmatrix} k \\ 2 \end{bmatrix}_q$ , and the width  $k-2$  is then at least 11. The positive cardinality-four term needs degree at most two. Thus these ten deficits persist for every  $k \geq 13$ . Their necessity is established; sufficiency in all degrees is the unproved content of Conjecture 1.

## 3 An exact conditional route to the original problem

Set

$$E_k = \sum_{j \geq 1} q^{2j^2} \begin{bmatrix} k \\ 2j-1 \end{bmatrix}_q - \sum_{\ell \geq 1} q^{8\ell^2 - \ell} \begin{bmatrix} k \\ 4\ell-1 \end{bmatrix}_q.$$

All sums are finite under the zero convention.

**Lemma 1.** *For every  $k \geq 1$ ,*

$$D_{k+1} - D_k = q^{k+1} E_k, \quad E_{k+1} - E_k = q^{k+2} Z_k.$$

*Proof.* Use  $\begin{bmatrix} k+1 \\ s \end{bmatrix}_q - \begin{bmatrix} k \\ s \end{bmatrix}_q = q^{k+1-s} \begin{bmatrix} k \\ s-1 \end{bmatrix}_q$ . For the two summands of  $D$ , subtracting  $s$  from their exponents gives  $2j^2$  and  $8\ell^2 - \ell$ , respectively. For the positive summands of  $E$ , the resulting exponent is  $k+2+2j(j-1)$ , so replace  $j$  by  $j+1$ . For its negative summands it is  $k+2+8\ell^2-5\ell$ ; replacing  $\ell$  by  $\ell+1$  leaves  $8\ell^2+11\ell+3$ . Zero out-of-range terms account for both endpoints.  $\square$

The finite bridge used below consists of the following exactly enumerated facts:

$$\frac{D_k}{1-q} \succeq_q 0 \quad (1 \leq k \leq 17), \quad \frac{E_{17} - q^{19} B}{1-q} \succeq_q 0. \quad (8)$$

The bridge numerator has degree 162, minimum prefix zero, and final prefix 32873. Its prefixes at degrees 19, 20, 21 are zero. These facts were independently reconstructed by exhaustive subsets, not inferred from approximate sampling. Section 4.2 gives a short exact procedure for checking every coefficient and the terminal prefix.

**Proposition 1** (Conditional implication). *Conjecture 1, together with the finite checks (8), implies (1) for every  $k \geq 1$ .*

*Proof.* For  $K \geq 17$ , Lemma 1 gives

$$\begin{aligned} \frac{E_K}{1-q} &= \frac{E_{17}}{1-q} + \sum_{t=17}^{K-1} q^{t+2} \frac{Z_t}{1-q} \\ &\succeq_q \frac{E_{17}}{1-q} - B \sum_{t=17}^{K-1} q^{t+2} \succeq_q \frac{E_{17} - q^{19} B}{1-q} \succeq_q 0. \end{aligned}$$

The second inequality uses  $B \succeq_q 0$  and extends the finite geometric sum to its infinite formal series. Summing the first recurrence from 17 now gives  $D_K/(1-q) \succeq_q 0$  for every  $K \geq 17$ . The remaining cases are (8). Multiplication by  $\prod_{i=2}^K (1-q^i)^{-1}$  completes the implication.  $\square$

This proposition does not establish its uniform premise. In particular, the degree-162 calculation cannot by itself resolve the original problem.

### 3.1 Obstructions that motivated the conjecture

A direct monotonicity induction fails: exact enumeration gives

$$[q^{29}]D_{11}/(1-q) = 55, \quad [q^{29}]D_{12}/(1-q) = 54,$$

and  $[q^{17}]E_{11}/(1-q) = -1$ . Nevertheless the direct trace coefficients are

$$[q^{29}]F_{11} = 16821, \quad [q^{29}]F_{12} = 18066.$$

Even eventual monotonicity of the unsmoothed numerators fails:  $[q^{13}]E_k = -1$  for every  $k \geq 12$ . The contributions are  $1+5-7$ ; the relevant small rectangle coefficients are already stable. One-sided pairing also fails: at  $k = 22$  the coefficient of  $q^{45}$  in

$$\frac{q^4 \begin{bmatrix} k \\ 2 \end{bmatrix}_q - q^{11}(1-q) \begin{bmatrix} k \\ 4 \end{bmatrix}_q}{1-q}$$

is  $231 - 237 = -6$ . These are obstructions to particular sufficient routes, not counterexamples to the trace problem or to Conjecture 1. They motivate controlling a fixed negative part instead of assuming unqualified monotonicity.

## 4 Exact evidence and targeted falsification

The exploratory computations identified the ten-term defect and suggested the recurrences. This discovery stage is distinct from exact certification of individual finite statements. The stored producers use arbitrary-precision integer arrays for complete finite products; Gaussian reconstruction and exhaustive subset enumeration provide different checking algorithms. The exactness of the arithmetic does not turn a finite scan into a uniform proof, and agreement between implementations is not formal verification of their source code.

The archived complete-row scan covers  $1 \leq k \leq 1004$ . It records full-vector comparisons and prefix summaries, rather than only numerical plots. The independent conjecture audit reconstructed all vectors for  $k \leq 64$  using a new  $q$ -Pascal implementation. It also reran subset enumeration through  $k = 17$ , the finite bridge, and full Gaussian reconstructions at  $k = 89, 159, 202, 205, 396$ . These five values were selected by a fixed random seed from  $1, \dots, 400$  without replacement; they are reproducibility checks within the scanned range, not new evidence outside it. An archived Gaussian computation at  $k = 1001$  agrees with the subset producer. The accepted audit checked equality of the stored records for that row; it did not rerun that large Gaussian computation or regenerate all 1004 rows. No claim of independent double certification of the entire large range is made.

Targeted attacks mapped staircase subsets, rectangle centers, and complementary upper edges into legal  $(k, n)$  instances of (5). The archive contains 33097 mapped count records, of which 1103 were independently checked by enumeration. The probes include the first possible negative

term, adjacent degrees, rectangle midpoint and upper-edge neighbors, and the extremizer  $k = 13$ . Complete-row scans also include zero degree, empty subsets, missing negative terms, and every interior coefficient. Residue-filtered products supply a further structural check. Arbitrary sparse products and tensor powers were excluded: they need not correspond to a consecutive set  $\{1, \dots, k\}$  and so do not define counterexamples to this conjecture.

#### 4.1 Stabilization and the remaining corner

In the subset interpretation, new subsets in  $Z_{k+1} - Z_k$  contain  $k + 1$ . The smallest possible negative weight is  $k + 2$ , and the smallest positive weight is  $k + 3$ . Hence the difference vanishes below degree  $k + 2$ . It follows that  $[q^n]Z_k/(1 - q)$  is constant for  $k \geq n - 1$  (within  $k \geq 1$ ). The scan through 1004 therefore supports (5) for every  $k$  in degrees  $n \leq 1005$ . The unexcluded corner is precisely

$$k \geq 1005, \quad n \geq 1006.$$

The finite bridge propagates the tested  $Z_t$  rows through  $t = 1004$  to  $E_K/(1 - q) \succeq_q 0$  for  $17 \leq K \leq 1005$  and then to  $D_K/(1 - q) \succeq_q 0$  for  $K \leq 1006$ . Also

$$F_{k+1} - F_k = \frac{q^{k+1}(D_k + E_k)}{(q; q)_{k+1}}.$$

Here  $E_k$  begins with  $q^2$  and  $D_k$  has no terms below degree four, so the difference begins at degree  $k + 3$ . Thus the corresponding finite evidence for the original problem covers  $k \leq 1006$  in all degrees and all  $k$  in degrees  $n \leq 1008$ . Its unexcluded corner is  $k \geq 1007$ ,  $n \geq 1009$ . Both conclusions inherit the stated limits of the computational evidence.

#### 4.2 Reproducibility

The accompanying research files contain Python producers, exact JSON records, and independent checkers. From their common root, `computation/defect_scan.py` with `-last 1000` and a required `-output` path regenerates the first thousand rows; `computation/block03_construction_attacks.py` with `-last 1004` and a required `-output` path regenerates the extension and construction probes. `computation/verify_defect_selected.py` checks the five sampled Gaussian rows. `computation/verify_conditional_bridge.py` reconstructs (8) and sharpness by enumeration. The exact commands and data descriptions are in `certificates/README.md`. These files are retained with the research project; no public repository deposit is asserted here.

For a self-contained small-case verifier, enumerate all subsets  $S$  of  $\{1, \dots, k\}$  and initialize integer arrays  $d, e, z$  to zero. Writing  $s = |S|$  and  $w = \sum S$ , perform the following updates whenever the condition holds:

| Condition             | Array entry            | Increment |
|-----------------------|------------------------|-----------|
| $s \equiv 0 \pmod{2}$ | $d_{w+s/2}, z_{w+s/2}$ | +1        |
| $s \equiv 0 \pmod{4}$ | $d_{w+s/4}$            | -1        |
| $s \equiv 1 \pmod{2}$ | $e_{w+(s+1)/2}$        | +1        |
| $s \equiv 3 \pmod{4}$ | $e_{w+(s+1)/4}$        | -1        |
| $s \equiv 2 \pmod{4}$ | $z_{w+(s-2)/4}$        | -1        |

An array indexed through  $k^2 + 3k + 30$  is more than sufficient. Each update follows directly from the subset generating identity. Compute all cumulative sums, including leading zeros and the final

total. For sharpness use  $k = 13$  and compare the negative prefixes of  $z$  with (4). For (8), check  $d$  at every  $1 \leq k \leq 17$ , then subtract the ten coefficients of  $q^{19}B$  from  $e$  at  $k = 17$  and check its prefixes. This procedure enumerates at most  $2^{17}$  subsets per row and checks the complete finite assertions without a Gaussian recurrence. For large rows, multiply the two finite products in (6) by integer array updates and extract their specified residues; this uses  $O(k^3)$  integer additions for the products, whereas exhaustive enumeration visits  $2^k$  subsets.

## 5 Literature boundary and limitations

Han–Xiong’s equation (24) and Conjecture 6.2 already propose eventual coefficient positivity of the numerator denoted  $D_k$  here, for  $k \geq 19$  [3]. We do not claim that assertion as a new conjecture. The dated comparison supporting this note examined six sources, including the two trace papers above. The additional comparisons were as follows.

The hook-difference sums and positive transformations recalled by Berkovich–Dhar [1, equations (1.1), (3.12)–(3.16)] do not directly identify the two weights in (3): at a common lower index  $s$ , their exponents differ by  $s/4 + 1/2$ . A common shift does not remove this difference; no suitable already-positive seed for their transformations was established. The truncated theta expression recalled by Ding–Sun [2, equation (1.14)] has a common infinite product denominator, whereas our bound concerns finite Gaussian rectangles and division only by  $1 - q$ . Positivity after multiplication by an infinite positive series does not imply positivity before multiplication. Wang–Krattenthaler’s Borwein product [6, equation (1.1)] uses two negative-factor residue classes modulo three; it is not either of the positive-factor products in (6). Reiner–Stanton’s centered Gaussian differences [5, Theorems 1 and 5] have parity and centering hypotheses different from our shifted sum. Individual Gaussian symmetry and unimodality alone do not imply its compensated-prefix positivity.

Two further works illustrate related, but mathematically separate, coefficient-control mechanisms. Zeng obtains coefficient positivity for a family of Links–Gould polynomials from an explicit nonnegative factorization [8]; in another setting, a factorization of the difference generating function for a permutation probability is combined with coefficient estimates and certified finite checks to prove strict monotonicity from  $n = 30$  onward [7]. Neither invariant is a fractional Gaussian trace, and neither result settles (1) or Conjecture 1.

These comparisons exclude the stated direct applications only. They do not exhaust transformations, establish global novelty, or prove that no equivalent conjecture exists elsewhere. The comparison date and versioned sources fix a bounded literature claim, not a perpetual open-status claim.

No uniform proof or counterexample is known within this work, for either Conjecture 1 or (1). Auxiliary transfer bounds and two-neighbor pairing premises explored in the research files remain unproved and are not additional accepted conjectures of this note. The finite probes provide no estimates in the unbounded corner, no probabilistic confidence level, and no all-parameter formal proof. The independent audit was an assessment of a conjecture and its evidence, not journal peer review or acceptance.

The most direct next falsification test starts at  $k = 1005$  and examines  $n \geq 1006$ , computing the exact prefix in (5). A single negative integer, with its  $(k, n)$  and independently regenerated array, falsifies the conjecture. Such a failure would eliminate this fixed- $B$  sufficient route while leaving (1) open. To falsify the original comparison one must separately compute a negative coefficient of  $F_k$ , preferably directly from the Puiseux products defining the traces. Conversely a proof of the uniform defect bound, together with the finite bridge, would settle the specified  $r = 1/4$  family and give its sharp auxiliary defect control.

## Computational and language-model assistance

OpenAI Codex language-model assistance was used in the research workflow for proposing identities and conjectures, developing Python computations, searching and comparing literature, and drafting this manuscript. A separate language-model context performed the recorded independent conjecture audit and exact reconstructions on its stated ranges. Independence here means separation from the proposing context and, where specified, a distinct implementation; it does not mean independent human peer review. This manuscript was prepared in a separate source-preparation context using the accepted audit. Model output and confidence are not mathematical evidence. The supporting calculations use exact Python integer arithmetic; no floating-point computation certifies the inequalities. No Lean proof of the proposed bound is supplied. LaTeX compilation and page-by-page PDF inspection are production checks and remain logically separate from the mathematical evidence described here.

## References

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